6 Quantum single particle systems

In this section we see how to extend the treatment of a system in contact with a heat bath to quantum systems, which is straightforward. Because of the discrete nature of the energy levels for bound states in a quantum system we will see that describing an isolated system with conserved energy is more difficult.

We will only treat single particles in this section and treat quantum gases later. We thus will not need to address the issue of statistics here.

6.1 General aspects of quantum systems

The simplest quantum system may be described by its energy eigenvalues in constructing the canonical partition function.

Let the eigenvalues be E_n, where $n = 1, 2, \cdots, N$, where N may be infinite. (Since we are dealing with one particle there is no ambiguity in this use of N.) We will, for the moment, neglect the possibility of a continuum of states in addition, as occurs for ionised hydrogen, where the electron is unbound.

We may have degeneracy, i.e. several states with the same value of E. We will denote the number of states which have energy E_n by g_n, where g_n can be any positive integer. An example would be in the hydrogen atom where for the principal quantum number, n, being say 2, then there is one $\ell = 0$ state and three $\ell = 1$ states ($m_\ell = -1, 0, 1$) all multiplied by 2 for the two possible spin states. Then, labelling g_n by the principal quantum number we see $g_2 = 4 \times 2 = 8$. For $n = 1$, there is only the spin degeneracy, so $g_1 = 2$.

The counterpart of $W(E)$, often called the "statistical weight", is:

$$W(E) = \sum_{n=1}^{N} g_n \, \delta(E - E_n) \ .$$

We **cannot** immediately[1] define the entropy (for the microcanonical system, at fixed E) by $S(E) = k_{\text{B}} \ln W(E)$, as $\ln[\delta(E - E_n)]$ is a very strange (and ill-defined) quantity.

[1]The quantum microcanonical ensemble is defined by integrating over a small interval δE. So, in that sense, one could define $S(E) = \sum_{n=1}^{N} k_{\text{B}} \ln(g_n) \, \delta(E - E_n)$, implying (defining) $S(E) = 0$ if there is one state (or no states) in the interval δE. There is no *simple* counterpart to the ergodic hypothesis, so a picture of time averages does not carry over easily.

However defining the canonical partition function for a system in contact with a heat bath is straightforward. We start from the general definition which can encompass the quantum case:

$$\begin{aligned} Z(T,V,N=1) &= \int \mathrm{d}E\; W(E)\mathrm{e}^{-\beta E} \\ &= \sum_{n=1}^{N} g_n \int \mathrm{d}E\; \delta(E-E_n)\,\mathrm{e}^{-\beta E} \\ &= \sum_{n=1}^{N} g_n\,\mathrm{e}^{-\beta E_n}\ . \end{aligned} \tag{6.1}$$

This result is what one would expect.

6.2 Examples of degeneracy

Some simple – and not so simple – examples follow of degeneracy in energy spectra.

1. A particle of mass m in a one-dimensional box of size a has a non-degenerate spectrum:
$$E_n = \frac{\hbar^2}{2m}\left(\frac{\pi n}{a}\right)^2 \quad \text{and} \quad g_n = 1\ .$$
2. A one-dimensional simple harmonic oscillator with natural frequency ω has a non-degenerate spectrum:
$$E_n = (n+\tfrac{1}{2})\hbar\omega \quad \text{and} \quad g_n = 1\ .$$
3. A two-dimensional harmonic oscillator may be degenerate if the frequencies are the same in the x- and y-directions, ω. There are two quantum numbers, n_x and n_y with the energy being:
$$E_{n_x,n_y} = (n_x+\tfrac{1}{2})\hbar\omega + (n_y+\tfrac{1}{2})\hbar\omega = (n_x+n_y+1)\hbar\omega\ ,$$
So the states which have the same value of $n_x+n_y=N$ are degenerate. There are $N+1$ such states:
$$(n_x,n_y) = (N,0),(N-1,1),\cdots(1,N-1),(0,N)\ .$$
We may thus write the states as
$$E_N = (N+1)\hbar\omega \quad \text{and} \quad g_N = N+1\ .$$
4. For the hydrogen atom, the angular momentum quantum number obeys: $0 \le \ell \le n-1$, where the principal quantum number is n. Each value of ℓ has $2\ell+1$ states. So the total number of states associated with n (including a factor of 2 for spin degeneracy) is:
$$\begin{aligned} g_n &= 2\sum_{\ell=0}^{n-1}(2\ell+1) \\ &= 2\left(2\times\tfrac{1}{2}n(n-1)+n\right) \\ &= 2n^2\ . \end{aligned}$$

E.g. for $n = 2$, we have one $\ell = 0$ state and three $\ell = 1$ states ($m_\ell = -1, 0, +1$), i.e. four spatial states times 2 = 8 states. And $2 \times 2^2 = 8$ consistently.

5. Problems start to occur with a particle in a two-dimensional box where both sides are, say, a. The energy levels are

$$\begin{aligned} E_{n_x,n_y} &= \frac{\hbar^2}{2m}\left(\frac{\pi n_x}{a}\right)^2 + \frac{\hbar^2}{2m}\left(\frac{\pi n_y}{a}\right)^2 \\ &= \frac{\hbar^2\pi^2}{2ma^2}\left(n_x^2 + n_y^2\right) \ . \end{aligned}$$

Thus degeneracy hinges of the number of integers n_x, n_y, n'_x, n'_y such that:

$$\begin{aligned} \frac{\hbar^2\pi^2}{2ma^2}(n_x^2 + n_y^2) &= \frac{\hbar^2\pi^2}{2ma^2}(n_x'^2 + n_y'^2) \\ \Rightarrow \quad n_x^2 + n_y^2 &= n_x'^2 + n_y'^2 \ . \end{aligned}$$

The number of solutions to this is a problem in number theory, as is the counterpart in three dimensions.

6.3 States versus levels

So far we have discussed energy levels, E_n, and their degeneracies, g_n. Then as observed above, we may write for the statistical weight, $W(E)$:

$$W(E) = \sum_{n=1}^{N} g_n \delta(E - E_n) \ .$$

However sometimes (see end of this section) we may not want to sum over *levels* but the *states* themselves. This distinction is illustrated in Fig. (6.1) where the ten states of the hydrogen atom with the lowest energy are used.

Each level (in the case of the hydrogen atom) has in several states associated with it. Each state is unique. The number of states, in this case, is

$$N_{\text{states}} = \sum_{n=1}^{2} g_n = g_1 + g_2 = 2 + 8 = 10 \ .$$

We may then choose to write the statistical weight, $W(E)$, as either a sum over levels (n labelling levels, with $1 \le n \le 2$), or over states (i labelling states, with $1 \le i \le 10$):

$$W_{\text{hyd}}(E) = \sum_{n=1}^{2} g_n \delta(E - E_n) = \sum_{i=1}^{10} \delta(E - E_i)$$

$$\text{and in general} \quad W(E) = \sum_{n=1}^{N} g_n \delta(E - E_n) = \sum_{i=1}^{N_{\text{states}}} \delta(E - E_i)$$

$$\text{where} \quad N_{\text{states}} = \sum_{n=1}^{N} g_n \ .$$

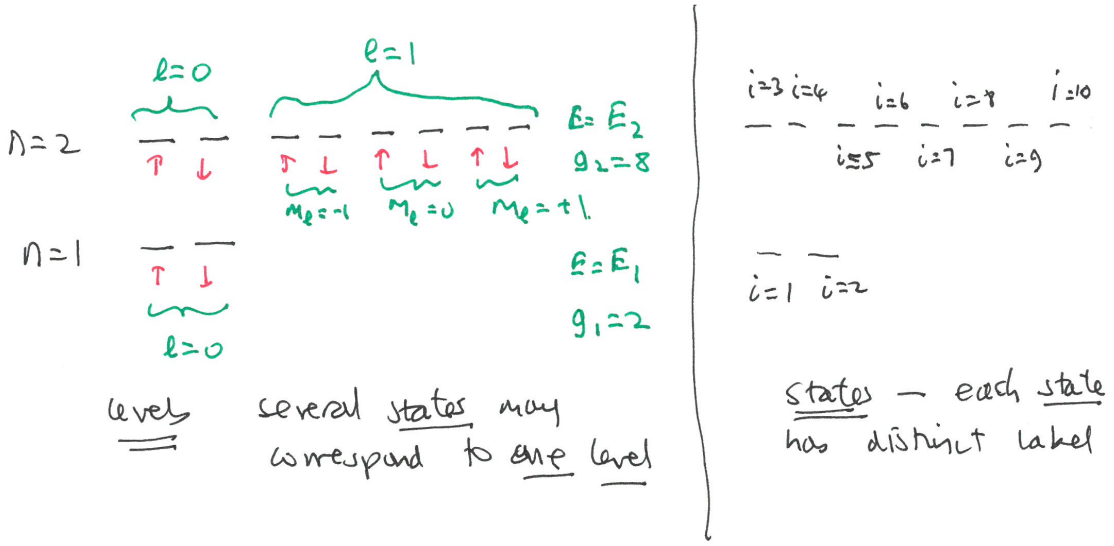

Figure 6.1: The lowest ten states of the hydrogen atom. **Left:** grouped into two energy **levels**, $n = 1$ and $n = 2$, labelled by the principal quantum number, n. The degeneracy of the $n = 1$ level is $g_1 = 2$, and that of the $n = 2$ level is $g_2 = 8$. **Right:** each **state** labelled by an integer $i = 1, 2, \cdots, 10$.

In the specific case of hydrogen, one might divide the sum over i into n, ℓ, m_ℓ and σ, the principal quantum number, the angular momentum ($\ell = 0$ or $\ell = 1$ here), the z-component of angular momentum (for $\ell = 1$, $m_\ell = -1, 0, +1$) and the spin, $\uparrow$ or $\downarrow$. This is overly complicated for the current purposes, but is physically precise.

6.4 Example: the simple harmonic oscillator

We work in one dimension for simplicity. Notice that we do not specify the volume as the potential confines the particle.

$$
\begin{aligned}
W_{\text{SHO}}(E) &= \sum_{n=0}^{\infty} \delta(E - [n + \tfrac{1}{2}]\hbar\omega) \\
Z(T, N = 1) &= \int \mathrm{d}E \; W_{\text{SHO}}(E) \mathrm{e}^{-\beta E} \\
&= \sum_{n=0}^{\infty} \mathrm{e}^{-\beta(n+1/2)\hbar\omega} \\
&= \mathrm{e}^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} \left(\mathrm{e}^{-\beta\hbar\omega}\right)^n \\
&= \frac{\mathrm{e}^{-\beta\hbar\omega/2}}{1 - \mathrm{e}^{-\beta\hbar\omega}} \\
&\underset{T\to\infty}{\sim} \frac{1}{\beta\hbar\omega} = \frac{k_{\text{B}}T}{\hbar\omega} \; .
\end{aligned}
$$

Note this is the same as the classical expression, Eqn. (3.7), in section 3.

We then may write down the Helmholtz free energy:

$$\begin{aligned} F(T, N=1) &= -k_{\mathrm{B}} T \ln Z(T, N=1) \\ &= \left[\tfrac{1}{2}\hbar\omega + k_{\mathrm{B}} T \ln\left(1 - \mathrm{e}^{-\beta\hbar\omega}\right)\right] \ . \end{aligned}$$

The first term represents the contribution to U of the zero-point energy. Note it is a constant – setting the zero for F, U etc, but not influencing any simple partial derivatives of the free energy.[2]

We may now calculate the entropy of the SHO, noting as promised the constant term is irrelevant:

$$\begin{aligned} S &= -\frac{\partial F}{\partial T} \\ &= -k_{\mathrm{B}} \ln\left(1 - \mathrm{e}^{-\beta\hbar\omega}\right) - k_{\mathrm{B}} T \frac{1}{1 - \mathrm{e}^{-\beta\hbar\omega}} (-1)(-\hbar\omega)\mathrm{e}^{-\beta\hbar\omega} \underbrace{\frac{\mathrm{d}\beta}{\mathrm{d}T}}_{=-1/(k_{\mathrm{B}}T^2)} \\ &= k_{\mathrm{B}} \left[-\ln\left(1 - \mathrm{e}^{-\beta\hbar\omega}\right) + \frac{\hbar\omega}{k_{\mathrm{B}}T} \frac{\mathrm{e}^{-\beta\hbar\omega}}{1 - \mathrm{e}^{-\beta\hbar\omega}} \right] \ . \end{aligned}$$

Now consider the limits of $S(T)$ for low and high temperatures. Firstly, $T \to 0$ or $\beta \to \infty$, then:

$$\begin{aligned} S(T) &\underset{T\to 0}{\sim} k_{\mathrm{B}} \left[-\left(-\mathrm{e}^{-\beta\hbar\omega}\right) + \frac{\hbar\omega}{k_{\mathrm{B}}T} \mathrm{e}^{-\beta\hbar\omega} \right] \\ &\simeq k_{\mathrm{B}} \frac{\hbar\omega}{k_{\mathrm{B}}T} \mathrm{e}^{-\beta\hbar\omega} \ . \end{aligned}$$

Here we picked the term tending to zero most slowly as $T \to 0$. Nevertheless $S \to 0$ very rapidly due to the exponential. This is because all states but the ground state are unpopulated because of the disparity of the two energy scales $k_{\mathrm{B}}T \ll \hbar\omega$.

Turning to high temperature, $T \to \infty$, $\beta \to 0$:

$$\begin{aligned} \mathrm{e}^{-\beta\hbar\omega} &\underset{T\to\infty}{\sim} 1 - \beta\hbar\omega \\ \Rightarrow \quad S &\underset{T\to\infty}{\sim} k_{\mathrm{B}} \left[-\ln(1 - 1 + \beta\hbar\omega) + \frac{\hbar\omega}{k_{\mathrm{B}}T} \frac{1 - \beta\hbar\omega}{\beta\hbar\omega} \right] \\ &\simeq k_{\mathrm{B}} \left[\ln\left(\frac{k_{\mathrm{B}}T}{\hbar\omega}\right) + 1 - \frac{\hbar\omega}{k_{\mathrm{B}}T} + \cdots \right] \end{aligned}$$

The largest term has a simple interpretation: the number of states accessible at temperature T is roughly

$$\begin{aligned} \text{“}W(k_{\mathrm{B}}T)\text{”} &= k_{\mathrm{B}}T/(\hbar\omega) \\ \Rightarrow \quad S(T) &\simeq k_{\mathrm{B}} \ln W(k_{\mathrm{B}}T) \\ &= k_{\mathrm{B}} \ln\left(\frac{k_{\mathrm{B}}T}{\hbar\omega}\right) \ , \end{aligned}$$

[2] But if one wishes to vary the spring constant (and hence frequency) of the oscillator to have a counterpart to changing the volume, then it would contribute to the counterpart to the pressure, "$(\partial F/\partial\omega)_T$".

as derived above.

The quantum harmonic oscillator is unusual as having a readily calculable Z and hence F. The particle in a box leads to *Theta functions*, see Eqn (42) for a general temperature. Sometimes one may deduce the high temperature behaviour using the Euler-Maclaurin summation formula.

6.5 Gibbs' entropy – liberation from equilibrium?

We will now derive an expression for entropy which can be used to define entropy in non-equilibrium systems and in more general contexts where one has a probability distribution. We mention it now as it is most easily *applied* to quantum systems, although we start the derivation for the classical case.

The derivation is **non-examinable**.

We will work within the canonical ensemble and make a comment at the end of this subsection about the applicability to the microcanonical case, where energy is fixed. The normalised probability distribution for one state of energy E is:

$$p(E) = \frac{\mathrm{e}^{-\beta E}}{Z(T,N)} \ ,$$

i.e. with the partition function, $Z(T,N)$, representing the normalisation of the Boltzmann factor.

Using the thermodynamic expression for entropy:

$$\begin{aligned} S = -\left(\frac{\partial F}{\partial T}\right)_V &= -\frac{\partial}{\partial T}\left(-k_{\mathrm{B}} T \ln Z\right) \\ &= k_{\mathrm{B}} \ln Z + k_{\mathrm{B}} T \frac{1}{Z}\left(\frac{\partial Z}{\partial T}\right)_V \ . \end{aligned} \tag{6.2}$$

Let us evaluate $(\partial Z/\partial T)_V$, suppressing the explicit statement about constant V for simplicity

$$\begin{aligned} \frac{\partial Z}{\partial T} &= \int \mathrm{d}E \ W(E) \frac{\partial}{\partial T} \mathrm{e}^{-\beta E} \\ &= \int \mathrm{d}E \ W(E)(-E)\mathrm{e}^{-\beta E} \underbrace{\frac{\mathrm{d}\beta}{\mathrm{d}T}}_{=-1/(k_{\mathrm{B}}T^2)} \\ &= \frac{1}{k_{\mathrm{B}}T^2} \int \mathrm{d}E \ W(E) E \mathrm{e}^{-\beta E} \ . \end{aligned} \tag{6.3}$$

Now substituting Eq. (6.3) into Eq. (6.2), we see:

$$
\begin{aligned}
S &= k_{\mathrm{B}} \ln Z + \frac{k_{\mathrm{B}} T}{k_{\mathrm{B}} T^2} \frac{1}{Z} \int \mathrm{d}E \; W(E) E \mathrm{e}^{-\beta E} \\
&= k_{\mathrm{B}} \left[\underbrace{\frac{Z}{Z}}_{=1} \ln Z + \beta \int \mathrm{d}E \; W(E) E \frac{\mathrm{e}^{-\beta E}}{Z} \right] \\
\text{(Write } Z \text{ in numerator explicitly.)} \quad &= k_{\mathrm{B}} \int \mathrm{d}E \; W(E) \frac{\mathrm{e}^{-\beta E}}{Z} \left[\ln Z + \beta E \right] \\
&= -k_{\mathrm{B}} \int \mathrm{d}E \; W(E) \frac{\mathrm{e}^{-\beta E}}{Z} (-1) \left[\ln Z + \beta E \right] \\
&= -k_{\mathrm{B}} \int \mathrm{d}E \; W(E) \frac{\mathrm{e}^{-\beta E}}{Z} \ln \left[\frac{\mathrm{e}^{-\beta E}}{Z} \right] \\
\left(\text{Use } p(E) = \frac{\mathrm{e}^{-\beta E}}{Z}. \right) \quad &= -k_{\mathrm{B}} \int \mathrm{d}E \; W(E) p(E) \ln p(E) \; . \qquad (6.4)
\end{aligned}
$$

Now let us use this for the quantised case where the E's are discrete, E_n, with degeneracies g_n, i.e. energy level n includes g_n states. Then:

$$
W(E) = \sum_{n=1}^{N} g_n \delta(E - E_n) \; ,
$$

and upon substitution into Eq. (6.4), and using the delta functions to perform the E-integral, we see:

$$
\begin{aligned}
S = -k_{\mathrm{B}} \int \mathrm{d}E \; W(E) p(E) \ln p(E) &= -k_{\mathrm{B}} \int \mathrm{d}E \; \sum_{n=1}^{N} g_n \delta(E - E_n) p(E) \ln p(E) \\
&= -k_{\mathrm{B}} \sum_{n=1}^{N} g_n \; p(E_n) \ln p(E_n) \; . \qquad (6.5)
\end{aligned}
$$

This expression is normally quoted as a **sum over states** not energy levels i.e. without the g_n as the sum runs over the bigger range,

$$
N_{\text{states}} = \sum_{n=1}^{N} g_n \geq N \; ,
$$

with equality only if all $g_n = 1$. Then:

$$
S = -k_{\mathrm{B}} \sum_{i=1}^{N_{\text{states}}} p_i \ln p_i \; , \qquad (6.6)
$$

which is the form quoted most usually.

6.5.1 Microcanonical Gibbs entropy

For the case of fixed energy, i.e. the microcanonical case, the probabilities, p_i, of each of the $W(E)$ states which have energy E are equal ("equal a priori probabilities"), hence are

$$p_i = \frac{1}{W(E)} \text{ , implying, correctly } \quad \sum_{i=1}^{W(E)} \frac{1}{W(E)} = \frac{1}{W(E)} \quad \sum_{i=1}^{W(E)} 1 = \frac{1}{W(E)} W(E) = 1 \ ,$$

i.e. correct normalisation.

Then:

$$\begin{aligned} S &= -k_{\text{B}} \sum_{i=1}^{W(E)} p_i \ln p_i \\ &= -k_{\text{B}} \sum_{i=1}^{W(E)} \frac{1}{W(E)} \ln \left(\frac{1}{W(E)} \right) \\ \text{(Note summand independent of } i.\text{)} \quad &= k_{\text{B}} \frac{1}{W(E)} \ln W(E) \underbrace{\sum_{i=1}^{W(E)} 1}_{=W(E)} \\ &= k_{\text{B}} \frac{1}{W(E)} \left(\ln W(E) \right) W(E) \\ &= k_{\text{B}} \ln W(E) \ , \end{aligned} \tag{6.7}$$

in agreement with the Boltzmann expression.

The advantage of the Gibbs expression for the entropy is to use it to define S when one does not have the equilibrium thermodynamic framework. For example one can use it in a non-equilibrium context (although one needs some other source of the values of p_i), or one can use it to define the "information entropy" – see problem sheet 3.

6.6 Interpretation of First Law in quantum terms

Here we construct an expression of the First Law of thermodynamics

$$\mathrm{d}U = T\mathrm{d}S - p\,\mathrm{d}V \tag{6.8}$$

for a quantum system in terms of the changes of energy *states*, E_n (the variable n may denote more than one number, e.g. n_x, n_y and n_z for a three-dimensional box), and their probabilities, p_n. (Note here we do not use levels, so no degeneracy factors.)

Firstly note, taking differential of p_n:

$$p_n = \frac{\mathrm{e}^{-\beta E_n}}{Z} \quad \text{and} \quad \sum_n p_n = 1 \quad \Rightarrow \quad \sum_n \mathrm{d}p_n = 0 \ , \tag{6.9}$$

where Z is the canonical partition function and the sums run over *states*.

Then

$$U = \sum_n E_n p_n$$

$$\Rightarrow \quad \mathrm{d}U = \sum_n \left(E_n \mathrm{d}p_n + p_n \mathrm{d}E_n\right) \tag{6.10}$$

$$\text{total differential} \quad \mathrm{d}U = \sum_n \left[\left(\frac{\partial U}{\partial p_n}\right)_{E_n} \mathrm{d}p_n + \left(\frac{\partial U}{\partial E_n}\right)_{p_n} \mathrm{d}E_n\right] \tag{6.11}$$

$$\text{(Equating differentials)} \quad E_n = \left(\frac{\partial U}{\partial p_n}\right)_{E_n} \quad \text{and} \quad p_n = \left(\frac{\partial U}{\partial E_n}\right)_{p_n} \ . \tag{6.12}$$

We use the Gibbs expression for the entropy, S,

$$S = -k_{\mathrm{B}} \sum_n p_n \ln p_n$$

$$\Rightarrow \quad \mathrm{d}S = -k_{\mathrm{B}} \sum_n \left[(\mathrm{d}p_n) \ln p_n + p_n \frac{1}{p_n} \mathrm{d}p_n\right]$$

$$\text{(Use Eq. (6.9))} \quad = -k_{\mathrm{B}} \sum_n \left[-\beta E_n \mathrm{d}p_n\right] - k_{\mathrm{B}} \underbrace{\sum_n \mathrm{d}p_n \left(-\ln Z \ + 1\right)}_{=0 \text{ as } \sum \mathrm{d}p_n = 0}$$

$$= \frac{1}{T} \sum_n E_n \mathrm{d}p_n$$

$$\Rightarrow \quad T\mathrm{d}S = \sum_n E_n \mathrm{d}p_n \ . \tag{6.13}$$

Substituting Eq. (6.13) into Eq. (6.10) we see

$$\mathrm{d}U = \sum_n \left(E_n \mathrm{d}p_n + p_n \mathrm{d}E_n\right)$$

$$= T\mathrm{d}S + \sum_n p_n \mathrm{d}E_n \ . \tag{6.14}$$

The energy of a state only depends on the volume, $E_n(V)$, through the allowed states in the box. E.g. in three dimensional cubic box, with $L = V^{1/3}$, with quantum numbers n_x, n_y and n_z,

$$E_{n_x,n_y,n_z}(V) = \frac{\hbar^2 \pi^2}{2mL^2} \left(n_x^2 + n_y^2 + n_z^2\right)$$

$$= \frac{\hbar^2 \pi^2}{2mV^{2/3}} \left(n_x^2 + n_y^2 + n_z^2\right) \ . \tag{6.15}$$

Then

$$\sum_n p_n \mathrm{d}E_n = \sum_n p_n \frac{\mathrm{d}E_n}{\mathrm{d}V}\mathrm{d}V \ .$$

Remembering Eq. (6.12), that the $\{pn\}$ are fixed, we may pull the V derivative through the expression while it becoming a partial derivative:

$$\begin{aligned}\sum_n p_n \mathrm{d}E_n &= \left(\frac{\partial}{\partial V}\sum_n p_n E_n\right)_{p_n} \mathrm{d}V \\ &= \left(\frac{\partial \langle E\rangle}{\partial V}\right)_{\{p_n\}} \mathrm{d}V \ . \end{aligned} \tag{6.16}$$

But since

$$S = -k_{\mathrm{B}} \sum_n p_n \ln p_n \ ,$$

we see if all $\{p_n\}$ are fixed, S if fixed so:

$$\begin{aligned}\sum_n p_n \mathrm{d}E_n &= \left(\frac{\partial \langle E\rangle}{\partial V}\right)_{\{p_n\}} \mathrm{d}V \\ &= \left(\frac{\partial U}{\partial V}\right)_{S} \mathrm{d}V \\ \sum_n p_n \mathrm{d}E_n &= -p\,\mathrm{d}V \ . \end{aligned} \tag{6.17}$$

We have identified the average value of the energy with the internal energy, $\langle E\rangle = U$, and then used the thermodynamic expression for the pressure, p. Substituting Eq. (6.17) into Eq. (6.14), we see

$$\mathrm{d}U = T\mathrm{d}S - p\,\mathrm{d}V \ ,$$

as desired.

What we have shown is that there is a natural separation between changes to energy levels being associated with work and changes in occupancies (probabilities) being associated with entropy (or heat).

The results of this subsection allow us to derive a form of the equation of state relating p, V and U. Remembering Eq. (6.15), we see

$$\begin{aligned}\frac{\mathrm{d}E_n}{\mathrm{d}V} &= \frac{\mathrm{d}}{\mathrm{d}V}\frac{\hbar^2\pi^2}{2m}\frac{n_x^2+n_y^2+n_z^2}{V^{2/3}} \\ &= \frac{\hbar^2\pi^2}{2m}(-1)\frac{2}{3}\frac{n_x^2+n_y^2+n_z^2}{V^{5/3}} \\ &= -\frac{2}{3}\frac{1}{V}E_n \ . \end{aligned} \tag{6.18}$$

Consider the following from Eq. (6.17)

$$\begin{aligned} -p\,\mathrm{d}V &= \sum_n p_n \frac{\mathrm{d}E_n}{\mathrm{d}V}\mathrm{d}V \\ \text{(Use Eq. (6.18).)} \quad &= -\frac{2}{3}\frac{1}{V}\sum_n p_n E_n \mathrm{d}V \\ \text{(Use } \langle E\rangle = U.\text{)} \quad &= -\frac{2}{3}\frac{U}{V}\mathrm{d}V \\ \Rightarrow \quad pV &= \frac{2}{3}U \ . \end{aligned} \tag{6.19}$$

We will find similar results in the case of waves in the next section.

6.7 Expectation values of other quantum operators

We have shown how to determine the average value of the energy of a quantum system at temperature T. We now turn to other operators, for example the potential energy, $V(x)$.

This is discussed more formally in *Quantum Mechanics 3*. Here we merely say that noting the matrix elements of an operator, $\hat{O}$, in the eigenbasis of the Hamiltonian,

$$\mathcal{H}|n\rangle = E_n|n\rangle \ ,$$

are $\langle n|\hat{O}|n\rangle = O_{nn}$, which may be interpreted as the quantum average of $\hat{O}$ in the state $|n\rangle$, then the thermal average sums such quantum averages, weighted by the thermal probability of observation of state with energy E_n,

$$\langle\hat{O}\rangle(T) = \sum_{n=0}^{\infty}\langle n|\hat{O}|n\rangle\,\frac{\mathrm{e}^{-\beta E_n}}{Z} \ , \text{ where } Z = \sum_{n=0}^{\infty}\mathrm{e}^{-\beta E_n} \ .$$

In some treatments $\langle\hat{O}\rangle(T)$ is demoted $\langle\langle\hat{O}\rangle\rangle$, representing both the quantum and thermal average being taken.

Noting that $\mathrm{e}^{-\beta E_n}$ is diagonal in the basis $\{|n\rangle\}$, we may write down the expression in an arbitrary basis in terms of a trace using that basis set:

$$\langle\hat{O}\rangle(T) = \mathrm{Tr}\left\{\hat{O}\frac{\mathrm{e}^{-\beta\hat{\mathcal{H}}}}{Z}\right\} = \mathrm{Tr}\left\{\hat{O}\hat{\rho}(T)\right\} \ ,$$

where

$$\hat{\rho}(T) = \frac{\mathrm{e}^{-\beta\hat{\mathcal{H}}}}{Z} = \frac{\mathrm{e}^{-\beta\hat{\mathcal{H}}}}{\mathrm{Tr}\,\mathrm{e}^{-\beta\hat{\mathcal{H}}}}$$

is the thermal density operator (see *Quantum Mechanics 3*).

Example: Simple harmonic oscillator potential energy.
This is most readily performed using the ladder operators, i.e. using (the use of bras and kets is a little poetic here)

$$\begin{aligned}
\langle n|V(x)|n\rangle &= \frac{1}{2}K\langle n|x^2|n\rangle \\
&= \frac{1}{2}K\frac{\hbar}{2m\omega}\langle n|(a^\dagger + a)^2|n\rangle \\
&= \frac{1}{4}\hbar\sqrt{\frac{K}{m}}\langle n|a^\dagger a + aa^\dagger|n\rangle \\
&= \frac{1}{4}\hbar\omega\langle n|2\hat{N} + 1|n\rangle \\
&= \frac{1}{2}\hbar\omega(n + \tfrac{1}{2}) \\
&= \tfrac{1}{2}E_n \ ,
\end{aligned}$$

as expected (half of total energy is potential, on average). This may then be susbtituted into the sum with Boltzmann weights and that sum performed – eg by a "Feynman-style" trick,

$$\begin{aligned}
\langle V(x)\rangle(T) &= \frac{1}{Z}\sum_{n=0}^{\infty}\langle n|V(x)|n\rangle \mathrm{e}^{-\beta E_n} \\
&= \frac{1}{2Z}\hbar\omega\sum_{n=0}^{\infty}(n + \tfrac{1}{2})\mathrm{e}^{-\beta\hbar\omega\left(n+\frac{1}{2}\right)} \\
&= \frac{1}{2Z}\left(-\frac{\partial}{\partial\beta}\right)\underbrace{\sum_{n=0}^{\infty}\mathrm{e}^{-\beta\hbar\omega\left(n+\frac{1}{2}\right)}}_{=Z} \\
&= \frac{1}{2}\left(-\frac{\partial}{\partial\beta}\right)\ln Z \\
&= \frac{1}{2}\left(\frac{\partial}{\partial\beta}\right)\beta F \\
&= \frac{1}{2}U \quad \text{(using the Gibbs-Helmholtz equation, end section 4.1)} \ ,
\end{aligned}$$

i.e. half of the thermal average energy as expected.

6.8 Summary

In this section we covered:

- the difference between *states* and *levels*, the notion of degeneracy;
- the form of the partition function for a quantum system with discrete energy levels;
- the explicit example of the simple harmonic oscillator (SHO);

- the low and high T behaviour for the SHO;
- the Gibbs formula for the entropy;
- the relationship of changes in energy levels and the probabilities of observation to the First Law of Thermodynamics.